

Functional Annotation Report: PP_2671 / PedS2 (UniProt Q88JH8)

Organism: *Pseudomonas putida* (strain KT2440 / ATCC 47054 / DSM 6125), PSEPK **Locus tag:** PP_2671 | **Gene name:** *pedS2* | **UniProt:** Q88JH8 **EC:** 2.7.13.3 (histidine kinase) | **Length:** 424 aa | **KEGG:** ppu:PP_2671

1. Summary (Answer to the Research Question)

PP_2671 encodes **PedS2**, the **sensor histidine kinase** of the **PedS2/PedR2 two-component signal-transduction system** of *Pseudomonas putida* KT2440. Its primary function is to **sense the environmental availability of rare-earth elements (lanthanides, Ln³⁺)** and to transduce this signal, by ATP-dependent autophosphorylation on a conserved histidine and subsequent phosphotransfer to its cognate LuxR-type response regulator **PedR2 (PP_2672)**, into a transcriptional output. This system operates the so-called "**REE-switch**" that inversely regulates the two functionally redundant periplasmic pyrroloquinoline-quinone (PQQ)-dependent ethanol dehydrogenases — the Ca²⁺-dependent **PedE** and the lanthanide-dependent **PedH** — allowing the bacterium to match its periplasmic alcohol/aldehyde (volatile organic compound) oxidation machinery to metal availability. PedS2 is an **inner-membrane protein** that perceives its signal in the **periplasm** and executes catalysis (autokinase activity) in the **cytoplasm**.

The gene identity is unambiguous: Wehrmann et al. (2018) explicitly map the locus pair **PP_2671/PP_2672 to PedS2/PedR2**, matching the UniProt annotation (histidine kinase, EC 2.7.13.3, membrane, HAMP + HATPase domains).

2. Gene / Protein Identity Verification

Check	Result
Symbol matches description	✓ PP_2671 = <i>pedS2</i> ; UniProt "histidine kinase" matches the characterized sensor kinase PedS2
Organism correct	✓ <i>P. putida</i> KT2440 (PSEPK)
Domains align with literature	✓ HAMP domain + HATPase_C/His-kinase catalytic domain = canonical two-component sensor kinase
Literature specificity	✓ Direct primary-literature characterization of PP_2671 as PedS2 (Wehrmann et al., 2018)

There is **no ambiguity**: this locus is directly and specifically characterized in the primary literature.

3. Primary Molecular Function

3.1 Enzymatic reaction (EC 2.7.13.3)

PedS2 is a **protein-histidine kinase**. As a two-component sensor kinase it catalyzes:

1. **Autophosphorylation**: $\text{ATP} + \text{protein L-histidine} \rightarrow \text{ADP} + \text{protein N-phospho-L-histidine}$ (on the conserved His of the DHp/H-box), catalyzed by the C-terminal HATPase (GHKL-family) ATP-binding domain.
2. **Phosphotransfer**: the high-energy phospho-His donates the phosphoryl group to a conserved **aspartate** on the receiver domain of the response regulator **PedR2**.

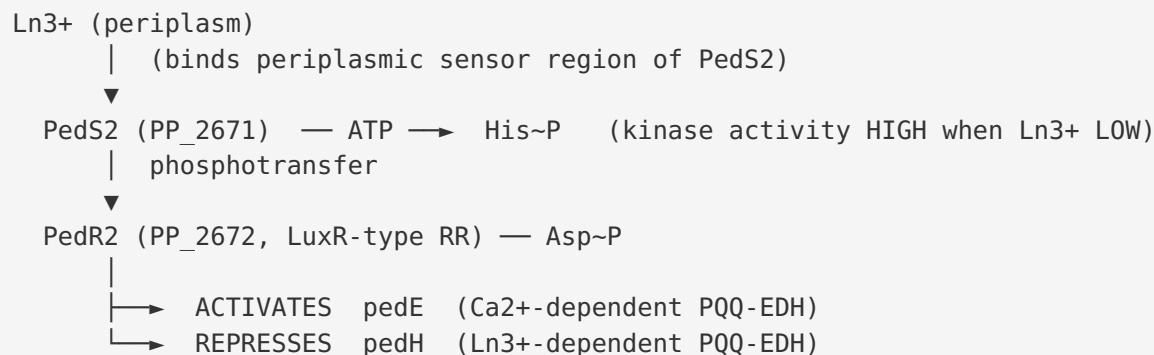
The "substrate" of the sensor kinase in the signaling sense is therefore its **cognate response regulator PedR2 (PP_2672)**; the chemical substrate/cofactor of the catalytic reaction is **ATP** (UniProt GO:0005524 ATP binding; GO:0000155 phosphorelay sensor kinase activity).

3.2 Input signal / ligand specificity

The physiological **input signal is the availability of lanthanide (Ln^{3+}) rare-earth elements** (experimentally probed with La^{3+}). Evidence (Wehrmann et al., 2018): - In the **absence of La^{3+}** , PedS2 is kinase-active $\rightarrow$ phosphorylates PedR2. - In the **presence of La^{3+}** , PedS2 kinase activity is **lowered**, proposed to occur through **direct binding of the metal ion to the periplasmic region of PedS2** (or an indirect interaction), reducing phospho-PedR2 levels.

PedS2 is described as a **putative La^{3+} -binding protein**; high iron can cause **mismetallation** of PedS2 and other La^{3+} -binding proteins, shifting the Ln^{3+} concentration required for switching (Wehrmann et al., 2019).

4. Signaling Pathway and Downstream Output ("the REE-switch")



- **Low/no lanthanide:** active PedS2 $\rightarrow$ phospho-PedR2 $\rightarrow$ **pedE ON, pedH OFF** $\rightarrow$ alcohol oxidation via the Ca^{2+} enzyme PedE.
- **Lanthanide present:** reduced PedS2 activity $\rightarrow$ less phospho-PedR2 $\rightarrow$ **pedE OFF, pedH de-repressed** (and additionally activated by a still-unknown regulatory module) $\rightarrow$ oxidation via the Ln^{3+} enzyme PedH.

This is the **first detailed molecular characterization of a rare-earth-element switch in a non-methylotrophic bacterium** (Wehrmann et al., 2018). PedR2 has a **dual regulatory role** (activator of *pedE*, repressor of *pedH*).

Broader pathway context: PedE and PedH are the terminal sensing/redundant enzymes of a **periplasmic oxidation system** that oxidizes a broad range of alcohols and aldehydes (volatile organic compounds such as ethanol and 2-phenylethanol), important for detoxification and catabolism (Wehrmann et al., 2017). Cytoplasmic uptake of lanthanides (which tunes the switch and enables PedH-dependent growth at low REE concentrations) is mediated by the **ABC transporter PedA1A2BC**, encoded in the same *ped* gene cluster (Wehrmann et al., 2019). Thus PedS2 is the regulatory sensor at the head of an integrated periplasmic C1/alcohol-oxidation and REE-adaptation module.

5. Subcellular Localization

- **UniProt:** Membrane.
 - **Bioinformatic topology (this work, Kyte-Doolittle hydropathy, window 19 on the 424-aa sequence):** an N-terminal transmembrane helix (~residues 17–30) and a second hydrophobic/TM segment (~residues 147–160) flank a ~117-residue **periplasmic sensor loop (~30–147)**; these are followed by the cytoplasmic **HAMP linker (161–213)** and the cytoplasmic **histidine-kinase catalytic domain (236–424)**.
 - **Interpretation:** PedS2 is an **integral inner-membrane** sensor kinase. It **perceives the Ln^{3+} signal in the periplasm** and **carries out autophosphorylation/phosphotransfer in the cytoplasm**, with the HAMP domain transmitting the conformational signal across the membrane. This topology is fully consistent with the experimental proposal that La^{3+} binds the periplasmic region.
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6. Domain Architecture (evidence from structure/sequence)

Region (aa)	Feature	Role
~17–30	TM helix 1	Membrane anchoring
~30–147	Periplasmic loop	Putative lanthanide (Ln ³⁺) sensing / ligand binding
~147–160	TM helix 2	Membrane anchoring
161–213	HAMP domain (IPR003660)	Transmembrane signal relay to the kinase core
236–424	Histidine-kinase catalytic core: DHp (His phosphoacceptor) + HATPase_C/GHKL ATP-binding (IPR003594, IPR036890, IPR005467)	Autokinase + phosphotransfer; ATP binding; dimerization (GO:0046983)

Family classification (InterPro IPR050482): **Sensor histidine kinase of two-component systems.**

6.1 Conserved catalytic residues (bioinformatic evidence, this work)

Sequence analysis of the cytoplasmic kinase core confirms all motifs required for a catalytically competent HisKA/GHKL kinase:

Motif	Position	Sequence context	Role
H-box (phosphoacceptor His)	His238	L-A-Q-T-L-H-D-D-L-G	Site of ATP-dependent autophosphorylation
N-box	~res 300–330	—	Coordinates Mg ²⁺ /ATP
G1 / D-box (DxGxG)	375–379	DNGCG	ATP phosphate binding (Bergerat fold)
G2 / G-box	411–413	GGG	ATP-lid / nucleotide binding

The intact H-box His238 and GHKL ATP-binding boxes provide independent structural evidence that PedS2 autophosphorylates on **His238** and then transfers phosphate to the Asp of PedR2 — consistent with the experimentally demonstrated PedR2 phosphorylation.

6.2 Genomic context (the *ped* cluster; UniProt survey PP_2662–PP_2683)

PedS2 (PP_2671) sits within a self-contained periplasmic PQQ-dependent oxidation cluster:

Locus	Product	Role
PP_2671	PedS2 (this protein) — sensor histidine kinase	Lanthanide sensing / signal input
PP_2672	PedR2 / ExaE — LuxR-type response regulator (212 aa)	Cognate partner; activates <i>pedE</i> , represses <i>pedH</i>
PP_2674	PedE — Ca ²⁺ -dependent PQQ-ADH (EC 1.1.2.8, 631 aa)	Effector enzyme (low-REE)
PP_2679	PedH — lanthanide-dependent PQQ-ADH (595 aa)	Effector enzyme (high-REE)
PP_2680	AldB-II aldehyde dehydrogenase	Downstream aldehyde oxidation
PP_2681	PqqD2 (PQQ biosynthesis)	Cofactor supply
PP_2675 / PP_2676	Cytochrome c / periplasmic binding protein	Electron transfer / periplasmic function
PP_2662 / PP_2669 / PP_2667–2668	Porin / OMP / ABC transporter	Substrate/metal traffic
PP_2664 / PP_2665 (AgmR)	Second sensor kinase / DNA-binding response regulator	Candidate for the additional, still-unidentified <i>pedH</i> -activating module

This co-localization with the cognate response regulator, both effector dehydrogenases, PQQ-biosynthesis and electron-transfer genes pinpoints PedS2 as the dedicated regulatory sensor gating this periplasmic alcohol/aldehyde-oxidation pathway. (Note: UniProt labels PP_2672 "ExaE"; Wehrmann et al. 2018 name the same protein PedR2 — they are identical.)

7. Supported vs. Refuted Hypotheses

Supported - PP_2671 = PedS2, a two-component **sensor histidine kinase** (EC 2.7.13.3). (*UniProt; Wehrmann 2018*) - PedS2 phosphorylates the LuxR-type response regulator **PedR2 (PP_2672)**. (*Wehrmann 2018*) - PedS2 activity responds **inversely to lanthanide availability**; it is a putative La^{3+} -binding sensor. (*Wehrmann 2018, 2019*) - Output is the **REE-switch** inversely regulating *pedE* (Ca^{2+} -PQQ-EDH) and *pedH* (Ln^{3+} -PQQ-EDH). (*Wehrmann 2018*) - PedS2 is an **inner-membrane protein with a periplasmic sensing region**. (*UniProt; topology analysis; Wehrmann 2018*) - PedS2 retains all catalytic motifs (H-box His238, GHKL ATP-binding boxes) of a functional autokinase. (*sequence analysis, this work*) - PedS2 is co-clustered with PedR2/ExaE, PedE, PedH, PQQ-biosynthesis and a second TCS (PP_2664/AgmR). (*UniProt genomic survey*)

Refuted / ruled out - PP_2671 is **not** a metabolic/biosynthetic enzyme acting on a small-molecule substrate; it is a signal-transduction protein-kinase. - The gene symbol is **not** ambiguous in this organism — it is specifically the *ped* cluster sensor kinase, not a generic/orthologous histidine kinase from another species.

8. Evidence Quality and Limitations

- **Direct experimental evidence** (adaptive evolution, site-directed mutagenesis, transcriptional reporter fusions, complementation) supports the physiological role of PedS2 in the REE-switch (Wehrmann et al., 2018; genetic/physiological data 2019).
- The **exact molecular mechanism of lanthanide sensing** (direct La^{3+} binding to the periplasmic domain vs. indirect interaction) is **not yet definitively established** — the authors present it as a supported hypothesis. No purified-protein biochemistry or structure of PedS2 with bound Ln^{3+} has been reported here.
- Membrane topology and the conserved His/Asp phosphotransfer chemistry are inferred from **sequence/domain (bioinformatic) analysis** consistent with the two-component-kinase family, not from an experimental PedS2 structure.
- The additional activating module for *pedH* remains unidentified.

Future directions: purified-protein autokinase/phosphotransfer assays; identification of the La^{3+} -coordinating residues in the periplasmic domain (site-directed mutagenesis); structural determination; identification of the additional *pedH*-activating regulator.

8b. Physiological & Evolutionary Significance (context)

- **A conserved regulatory paradigm — the "lanthanide switch."** The inverse, REE-dependent regulation of Ca^{2+} - vs Ln^{3+} -utilizing PQQ dehydrogenases that PedS2 operates is a recurring theme across bacteria; it was first described in methylotrophs and is the subject of multiple authoritative reviews (Skovran et al. 2019, [P 31166187](#) Featherston & Cotruvo 2021, [P 32979423](#) Daumann et al. 2022, [P 36167440](#)). A mechanistic model for the switch in *Paracoccus denitrificans* was recently proposed (Parekh & Spiro 2025, [P 40846390](#)). The *P. putida* PedS2/PedR2 study represents the **first detailed characterization of this switch in a non-methylotroph** (Wehrmann et al. 2018).
- **Effector-enzyme substrate range (what the pathway ultimately does).** The PQQ-ADHs gated by PedS2 are broad-substrate periplasmic enzymes oxidizing volatile alcohols/ aldehydes — e.g. ethanol and 2-phenylethanol (Wehrmann et al. 2017, [P 28655819](#)) — and are of biotechnological interest (soluble, direct electron transfer, REE cofactor; PedE engineering: Wehrmann & Klebensberger 2018, [P 29239114](#)). Orthologues **IsPedE/IsPedH** in *Ideonella sakaiensis* participate in **ethylene-glycol**

metabolism, underscoring evolutionary conservation of the module (Hachisuka et al. 2022, [P 36289066](#)).

- **Note on the kinase's "substrate specificity."** PedS2 itself has **no small-molecule catalytic substrate**. Its specificity is (i) **ATP** for autophosphorylation and (ii) the **response regulator PedR2** for phosphotransfer, with **lanthanide ions acting as the modulatory input ligand**. The cellular La^{3+} response in *P. putida* is substrate-specific and also intersects glycerol metabolism (Wehrmann et al. 2020, [P 32345644](#)).

9. Key References

1. Wehrmann M, Berthelot C, Billard P, Klebensberger J. **The PedS2/PedR2 Two-Component System Is Crucial for the Rare Earth Element Switch in *Pseudomonas putida* KT2440.** *mSphere* 3(4):e00376-18 (2018).

[P 30158283](#)
— Direct identification of PP_2671/PP_2672 as PedS2/PedR2 and the sensing/phosphorylation mechanism.

2. Wehrmann M, Berthelot C, Billard P, Klebensberger J. **Rare Earth Element (REE)-Dependent Growth of *Pseudomonas putida* KT2440 Relies on the ABC-Transporter PedA1A2BC and Is Influenced by Iron Availability.** *Front Microbiol* 10:2494 (2019).

[P 31736923](#)
— PedS2 as a putative La^{3+} -binding protein; iron mismetallation; lanthanide uptake.

3. Wehrmann M, Billard P, Martin-Meriadec A, Zegeye A, Klebensberger J. **Functional Role of Lanthanides in Enzymatic Activity and Transcriptional Regulation of PQQ-Dependent Alcohol Dehydrogenases in *Pseudomonas putida* KT2440.** *mBio* 8(3):e00570-17 (2017).

[P 28655819](#)
— Discovery/characterization of PedE and PedH and the periplasmic PQQ-ADH oxidation system.

4. Wehrmann M, Klebensberger J. **Engineering thermal stability and solvent tolerance of the soluble quinoprotein PedE from *Pseudomonas putida* KT2440.** (2018).

P 29239114

— Confirms PedE = PP_2674, periplasmic PQQ-ADH; biotechnological relevance.

5. Wehrmann M, Toussaint M, Pfannstiel J, Billard P, Klebensberger J. **The Cellular Response to Lanthanum Is Substrate Specific and Reveals a Novel Route for Glycerol Metabolism in *Pseudomonas putida* KT2440.** (2020).

P 32345644

— Substrate-specific La³⁺ response.

6. Hachisuka S, et al. **Ethylene glycol metabolism in the PET-degrading bacterium *Ideonella sakaiensis*.** (2022).

P 36289066

— Conserved PedE/PedH orthologues (IsPedE/IsPedH) in ethylene-glycol metabolism.

7. Reviews on the lanthanide switch / lanthanome: Skovran et al. 2019

(**P** 31166187);

Featherston & Cotruvo 2021

(**P** 32979423);

Daumann et al. 2022

(**P** 36167440);

Parekh & Spiro 2025

(**P** 40846390).

8. UniProt Q88JH8; InterPro IPR050482, IPR003660 (HAMP), IPR003594/IPR036890 (HATPase), IPR005467 (His-kinase).